

- 14** Satellites A and B of masses m and $2m$ are placed in geostationary orbits of radii r_A and r_B about the Earth, where the radii are measured from the centre of the Earth to the respective satellites. Which of the following statements is correct?
- A** The radii r_A and r_B are the same.
 - B** Both satellites have the same centripetal force.
 - C** Both satellites have the same total energy.
 - D** Both satellites have the same gravitational force.

Ans: A

For geostationary orbit, $T = 24$ hours

Gravitational force provides centripetal force, $\frac{GMm}{r^2} = mrw^2$

$$w = \frac{2\pi}{T} \text{ so } \frac{GMm}{r^2} = mr\left(\frac{2\pi}{T}\right)^2$$

$\therefore T^2 \propto r^3$ so orbital radii are the same for all geostationary satellites, independent of mass.

Gravitational force, centripetal force and total energy are dependent on mass.